Ultrastructural calcium simulations in neurons

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Summary

In this project we intend to study the calcium dynamics in reconstructed neurons, where the endoplasmic reticulum (ER), a multifunctional intracellular organelle of a neuron that consists of a complex three-dimensional network of connected endomembrane tubules, stacks and cisternae, is involved in various ion exchange processes. Of interest here are the three-dimensional spatiotemporal Ca^{2+} and Inositol-3-Phosphate (IP_3) dynamics in the intracellular space of neurons. We model these biochemical dynamics by a system of diffusion-reaction equations with highly nonlinear flux-boundary conditions and the Poisson-Nernst-Planck (PNP) equations. Using finite volume discretization together with geometric multigrid methods (GMG), we intend to run systematic numerical simulations of the calcium model on (1) Spine-Dendrite geometries and (2) full cell geometries. While only 1d morphological neuron data is directly available, our project requires a detailed 3d representation of the neuronal computational domain. For this we developed an automated pipeline to generate the necessary domains. Thus, we (3) intend to study and reconstruct cells from the `www.neuromorpho.org` database using available HPC resources. In total, this project should positively impact the computational mathematics research community, as well as parallel computing and neuroscience fields.

1 Research Objectives

Computational Neuroscience is a rapidly developing research field that demands more and more computing resources for solving complex problems. Where in the past, models of signal processing in brain cells were strongly reduced – due to the lack of computing power and high resolution experimental data – they are now being developed in more detail. Emerging areas of research encompass large scale network simulations, three-dimensional simulations of neuronal signals (**?????**) and multiscale modeling and simulation in neuroscience (**?**).

In order to resolve biochemical and electrical signals of neurons, based on physical first principles, models defined by sets of highly nonlinear and coupled partial and ordinary differential equations need to be solved numerically and efficiently. Recently, biophysical models of electrical and biochemical signaling, in particular calcium dynamics, have been developed and implemented in the multiphysics simulation platform uG4 (**?**). uG4 has been shown to scale optimally on massively parallel architectures (**?**). In various research projects the interplay between three-dimensional cell and organelle architecture and calcium signaling could be demonstrated (??????). From these projects two currently active ones emerged that focus on two distinct cellular scales: (1) the ultrastructural dynamics at contact points between two neurons (synapses/synaptic spines) and (2) whole cell calcium dynamics including the calcium exchange mechanisms at the cellular plasma membrane and the endoplasmic reticulum. The importance of studying calcium dynamics are manifold. For instance, spine and cellular calcium dynamics are implicated in the long-term effects of repetitive transcranial magnetic stimulation (rTMS), a technique used in clinical settings to treat e.g. schizophrenia. At the whole cell level calcium-induced calcium release dynamics controlled by the endoplasmic reticulum have been shown critical in learning, memory formation, and cell survival (????), as well as in neurodegenerative diseases, such as Alzheimer's and Parkinson's disease (????).

The objectives in this research proposal are to enable high-throughput numerical simulations to study the effect of a multitude of geometric and biological parameters on calcium signaling pathways in neurons. An active NIH funded research project on multiscale modeling of rTMS, in collaboration with Dr. Opitz (Engineering, University of Minnesota), Dr. Jedlicka (Computational Neuroscience, University of Giessen, Germany), and Dr. Vlachos (Neurophysiology, University of Freiburg, Germany), as well as Ph.D. research focusses on the effect of synapse loss on synapse to cell nucleus communication will be supported by the computing resources applied for below. Resources will support the following two research objectives:

1. At the level of synapses, synaptic spine morphology has been shown to change over time (?????). Along with this change, the intracellular cell organization adapts to ensure spine to dendrite coupling and calcium homeostasis in the spine head (?). In the proposed research, spine geometries that have been reconstructed in the rTMS project mentioned above, will be used to run 3D ultrastructural calcium simulations while scanning a parameter space consisting of receptor and calcium pump densities in the plasma- and endoplasmic membrane. These systematic simulation runs will identify critical parameter sets at which switching of neuronal behavior can be observed. Input for these simulations is voltage data derived from rTMS-induced electrical cell activity, which then drive calcium-induced calcium signals relevant for synaptic plasticity, cell adaption, and survival.

2. Whole cell calcium dynamics will be run by employing a 3D cell reconstruction method (?) which uses 1D neuroanatomical data that is readily available from morphology databases such as NeuroMorpho.org (?). This technique will allow high-throughput reconstruction of a large number of neurons on which parallel computations of cellular calcium signaling can be performed. Again, scanning a parameter space consisting of geometric parameters (e.g. ER size) and biophysical parameters, such as receptor densities, will ideally produce a rigorous understanding of critical states at which neurons loose their ability to reliably transmit calcium signals to the cell nucleus (which then triggers pathological states).

2 Computational Methology

The simulations of spatio-temporal dynamics outlined in Sec. ?? will be carried out with the multiphysics simulation package uG4 (??). This decades-old project focusses on solving partial differential equations on unstructured grids and has been successfully used in applications from diffusion-advection systems, Navier-Stokes equations, and mechanics applications. In particular the code has been developed to run very well on highly parallel architectures (**??**). To document uG4's general scaling behavior, we performed benchmark scaling simulations to solve the second order elliptic partial differential equation $\Delta\Phi = 0$, the Laplace equation with Dirichlet boundary conditions. The PDE problem was solved on the unit cube with seven levels of regular grid refinement which contains 16,974,593 DoFs. The scaling is shown in Tab. **??** of **Section General scaling properties of uG4** in the *performance* document. For this proposal we will be making use of first-order finite volume discretization methods for the underlying systems of partial differential equations. The resulting system of linear equations will be solved using a geometric multigrid method (GMG), combined with an implicit time-stepping scheme. For more details we refer to Sec. **??**.

3 Application efficiency

The SDSC Comet CPU resource is suitable for our research because it allows capacity computing and a quick turnaround on small to medium-scaled jobs of shorter runtimes, which makes it ideal for running the parameter sweep simulations in Sec. **??**. The large shared memory per node (128 GiB) allows us to load large computational grids which would not be possible with the smaller shared memory per node on comparable resources. Comparing our demands with the XSEDE Resource Selector[1] confirmed our pre-selected resources. Application efficiency is documented in the attached file *performance*.

4 Computational Research Plan

4.1 Synaptic spine dynamics

The model: Synaptic spines are protrusions from dendrites and form contacts with chemically connected cells. Examples of 3D-reconstructed spines are shown in Fig. **??**. The spine-interior often contains an organelle, the spine apparatus (SA), which is a continuous extension of the endoplasmic reticulum (ER) in the dendrite. The ER is a multifunctional intracellular organelle of a neuron, which consists of a complex three-dimensional network of connected endomembrane tubules, stacks and cisternae, is involved in various ion exchange processes, **?**, and ultimately responsible for synaptic plasticity. The SA and ER membrane harbors calcium exchangers that operate between the cytosol and ER domains. Typically, these are IP3-receptors, Ryanodine-receptors, and SERCA pumps (see Fig. **??**A). On the plasma membrane additional calcium exchangers operate between the extra- and intracellular space, namely Plasma Membrane Calcium Pumps (PMCA) and sodium-calcium-exchangers (NCX). The relevant quantities that need to be tracked on the 4D space/time cylinder are intracellular calcium Ca^{2+}, the molecule IP_3, which activates IP3-receptors together with Ca^{2+}, and mobile buffers, such as calbindin-D28k. One can define the following diffusion-reaction system:

[1] Information was retrieved from `https://portal.xsede.org/allocations/resource-info`

The interaction between cytosolic Ca^{2+} calbindin-D_{28k} (CalB) is described by

$$Ca^{2+} + \text{CalB} \underset{\kappa_b^-}{\overset{\kappa_b^+}{\rightleftharpoons}} [\text{CalBCa}^{2+}]. \tag{1}$$

The full domain equations for cytosolic calcium and calbindin-D_{28k} are thus given by

$$\frac{\partial c_c}{\partial t} = \nabla \cdot (D_c \nabla c_c) + \left(\kappa_b^- \left(b^{\text{tot}} - b\right) - \kappa_b^+ \, b \, c_c\right), \tag{2}$$

$$\frac{\partial b}{\partial t} = \nabla \cdot (D_b \nabla b) + \left(\kappa_b^- \left(b^{\text{tot}} - b\right) - \kappa_b^+ \, b \, c_c\right) \tag{3}$$

Exponential IP_3 decay towards a basal IP_3 concentration p^r in the cytosolic space is modeled by a reaction term that is added to the IP_3 diffusion equation, leading to the diffusion-reaction equation

$$\frac{\partial p}{\partial t} = D_p \Delta p - \kappa_p (p - p^r) \tag{4}$$

for IP_3 in the cytosolic domain. Endoplasmic Ca^{2+} dynamics are modeled by simple diffusion

$$\frac{\partial c_e}{\partial t} = D_c \Delta c_e \tag{5}$$

in the endoplasmic domain. Nonlinear Neumann boundary conditions, which are defined by the calcium channel and pump models depicted in Fig. **??**, complete the model. For the specifics of the ordinary differential equations that define PMCA, NCX, RyR, and SERCA pumps we refer to **?**.
Expected outcome: Previous work using simplified, parametrically designed, geometries (**?**) demonstrated that the 3D organization of synaptic spines plays a critical role. We now hypothesize that spine morphologies fall into distinct functional clusters. To test this hypothesis will use real morphology data sets (see Fig. **??**) and run simulations with different model parameter sets to determine under which conditions calcium signals can be reliably transmitted to the dendrite.
By simulating calcium dynamics on a range of spine morphologies (these can vary in shape and size, see Fig. **??**) and varying the parameters of the model equations (see **?** for specific parameter sets), we will be able to analyze the influence of cell and organelle morphologies on the calcium coupling behavior between spine and dendrite, which will provide novel insight into long-term potentiation of synapses.

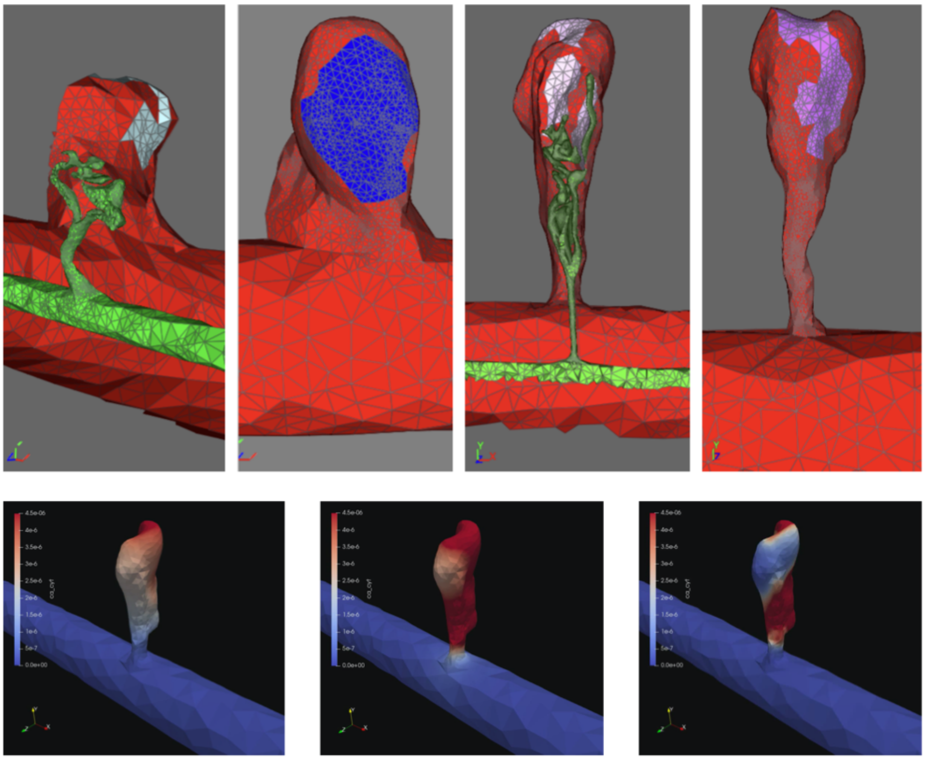


Figure 1: **Spine reconstructions and simulations:** (Top) Examples of spine reconstructions in collaboration with Dr. Vlachos (University of Freiburg, Germany). Reconstructions show variability in spine shape (red) as well as ER organization (green). (Bottom) Time sequence of a calcium simulation on reconstructed spine and dendrite geometry. Color scheme: red: max. calcium concentration, blue: min. calcium concentration

Numerical method: To computationally solve the model problem, the triangular surfaces meshes of synaptic spines (see Fig. **??** (C, D)), which have been generated from electron microscopy data (in collaboration with Dr. Vlachos, University of Freiburg, Germany) will be used to produce tetrahedral volume meshes using ProMesh (**?**). Using a first order finite volume method **?** the model equations are discretized on the previously generated surface and volume meshes. The nonlinear parts of the model, i.e. the Neumann boundary conditions are linearized with a Newton iteration. The resulting system of linear equations are solved by a geometric multigrid (GMG) method which is preconditioned using Bi-CGSTAB. The GMG solve of the linear system uses 3 iterations of pre- and post-smoothing, both of which use the *incomplete* LU factorization of the system matrix. We also set the *SuperLU* library as the base solver for our large, sparse, nonsymmetric matrix problem. A backward Euler method is used to solve the problem in time. The proposed numerical methods are available in uG4, which will be used for all proposed computations.
Estimate of proposed simulation runs: Next we describe the simulations we will perform for the above project and anticipated resources. The accuracy of the simulations relies on the following: employing physiologically correct geometries, running the simulations on geometries processed through several levels of grid refinements, running the simulations with small time step size Δt to capture the calcium dynamics in the cytosol and ER volume, and running the simulations with long end times. The studies we will perform are as follows:

1. `Baseline runs` – In order to study the physiological dynamics inside the dendrite we first run simulations without ER present to establish a baseline of calcium concentrations and general passive behavior of the calcium inside the dendrite geometry. We will run 10 simulations using a different calcium influx value with end time $t_f = 0.02\ s$ and $\Delta t = 0.5\ \mu s$ which equates to $\frac{t_f}{\Delta t} = 40{,}000$ time steps.

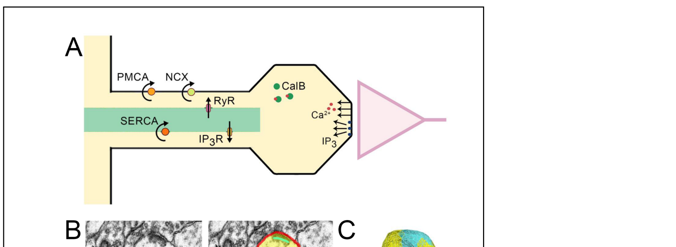
2. `RyR runs` – Next we run simulations with active ER having ryanodine receptors (RyR) and SERCA pumps activated. For these simulations we run a parameter sweep on the density of RyR on the ER, this will establish the critical RyR density for calcium to be released from the ER into the cytosol. We sweep the RyR density value from 0 to 10 RyR per μm^2 at increments of 0.01, this amounts to 1,000 simulation runs. These simulations have an end time of 20 milliseconds since coupling of calcium in the dendritic region of the cell has been observed during the first 20 milliseconds of spine activation. This simulation set also has 40,000 time steps with $\Delta t = 0.5\ \mu s$.

3. `Calcium Pulse runs` – Next we run studies for at least 3 seconds to observe the coupling effects between the spine region of the neuron and the dendrite region of the neuron for a fixed RyR density and different calcium influx pulse frequencies. These particular simulations will be performed with supercritical RyR density, each at 1 Hz, 3 Hz, 5 Hz, 10 Hz, and 20 Hz frequencies. This totals to 5 simulations, each with end time of 3 seconds, $\Delta t = 0.5\ \mu s$, this equates to $\frac{3.00}{0.5\times 10^{-6}} = 6{,}000{,}000$ time steps.

We have 9 different spine-dendrite geometries that will undergo each of the above simulations. Currently, with the XSEDE trial account, we ran these simulations with 0 refinement (20,605 Degrees of Freedom – DoFs) and 1 refinement (164,840 DoFs) on the Spine-Dendrite geometry and time step size $\Delta t = 0.5\ \mu s$. In Fig. **??** we show strong scaling for a geometry *without ER*. We ran a simulation with 164,840 DoFs across 384 cpu cores which took 4.69 minutes = 0.0782 hours with simulation end time 0.02 seconds. Therefore the number of core hours $= 384\ cores \times 0.0782\ hours = 30.02$ core-hours. In order to capture accurate results we will process geometries with at least 1 level of refinement, with and without ER geometry present. In Table **??** we indicate the anticipated compute time requirements for the sub-projects. The total number of runs is determined by (#runs/cell) × (#cells); for instance, the `Baseline runs` are 10 simulations and there are 9 cell geometries, therefore, we have (10 runs/cell) × (9 cells) = 90 runs. Then the number of core-hours are computed by (#runs) × (core-hours/run); therefore, for `Baseline runs` we have (90 runs) × (30 core-hours/run) = 2700 core-hours. Similarly for the `RyR runs` we have 2700 core-hours. For the `Calcium Pulse runs` we need to take into account the number of time steps since the end time is 3 seconds, we have $\frac{30 \text{ core-hours}}{40000 \text{ time-steps}} \times 6000000$ time-steps = 4500 core-hours.

Concerning storage requirements: we have two types of data output, text files in .txt format that contain the average calcium concentration in the volume and surface regions of the cell geometry and Visualization Toolkit files (vtk) for rendering and visualizing data in 3 dimensions. We will be generating .vtk output for the Calcium Pulse runs with active RyR & SERCA pumps and we anticipate that each vtk file is 0.24 GiB at 1 refinement level.

The Calcium Pulse simulations run for 3 seconds and total 6 million time steps but we will require only 100 vtk time samples (equally distributed in time)



out of the 6 million time steps. For the Calcium Pulse runs with active RyR & SERCA pumps we will require $\left(0.24\frac{\text{GiB}}{\text{file}}\right)\times(100\text{ files})\times(9\text{ cells})\times(5\text{ freq.}) =$ **1,080 GiB** of vtk storage for all nine cells and five frequencies. For the Calcium Baseline runs each simulation run will produce 2,750 kB (for end time of 0.02 seconds) of data in the form of text files, at 10 runs for 9 cells this equates to $\left(2750\frac{\text{kB}}{\text{run}}\right)\times(10\text{ runs})\times(9\text{ cells}) = 0.2475$ GiB. For the RyR & SERCA pump simulations we will be performing 1,000 simulation runs: $\left(2750\frac{\text{kB}}{\text{run}}\right)\times(1000\text{ runs})\times(9\text{ cells}) = 24.75$ GiB. For the Calcium pulse simulations we have a run time 3 seconds which equates to 412,500 kB per run; therefore, $412,500\times 15\times 9 = 55.6875$ GiB. Therefore, in total we anticipate $0.2474 + 24.75 + 55.6875 =$ **80.685 GiB** for text data; hence, for all spine simulations we anticipate $80.685 + 1,080 =$ 1.16 TiB. If we include ER geometry and dynamics this will incur more storage; therefore, we anticipate per month approximately **2 TiB** after which all data will be copied to local storage.

4.2 Whole cell dynamics

The model: While the first subproject in Sec. **??** focusses on the ultrastructure of individual synaptic spines, this subproject is defined on the whole cell level. This enables a rigorous analysis of how and when calcium signals are reliably relayed to the cell nucleus. The calcium model equations are the same as in Sec. **??**. The computational meshes are derived in the following way. The database NeuroMorpho.org (**?**) hosts 1D geometry data of neurons. These data contain vertex and edge information of the neuron graph. Additionally each vertex stores the dendrite radius. With these input data, the tool AnaMorph (**?**) will be used to generate surface meshes of the corresponding 1D geometry. The ER will be modeled as a neuron-within-neuron and can be generated alongside the surface reconstruction. In a final step the volumetric mesh is generated with ProMesh (**?**). The reconstruction pipeline is illustrated in Fig. **??**. Simulations will be run by initializing the calcium model with synapses. Synapses will be placed at distinct locations on the dendritic tree. Using standard synapse models calcium influx will be modeled as time-dependent Neumann boundary conditions.
Expected outcome: The ability for neurons to communicate calcium signals to the cell nucleus are critical for learning, plasticity, and cell survival (**????**). In neuropathological situations, the calcium pathways are disrupted (**????**).

We intend to study which conditions are necessary for neurons to reliably transmit calcium signals to the nucleus. We hypothesize that the spatio-temporal synaptic input patterns are relevant for synapse-to-nucleus communication. Thus, we intend to run simulations with different spatial synapse profiles and different temporal profiles, to test the influence of synapse clustering and synchronous vs. asynchronous activation. We further hypothesize that the morphological and biophysical parameters are important in stable communication. Thus, we plan to vary the dendrite to ER diameter ratios, as well as calcium channel and pump densities to determine threshold parameter sets that distinguish between stable and unstable synapse-to-nucleus communication. The results of this study could provide new insight into pharmacological intervention during e.g. synapse loss, which is a major contributor to Alzheimer's disease.

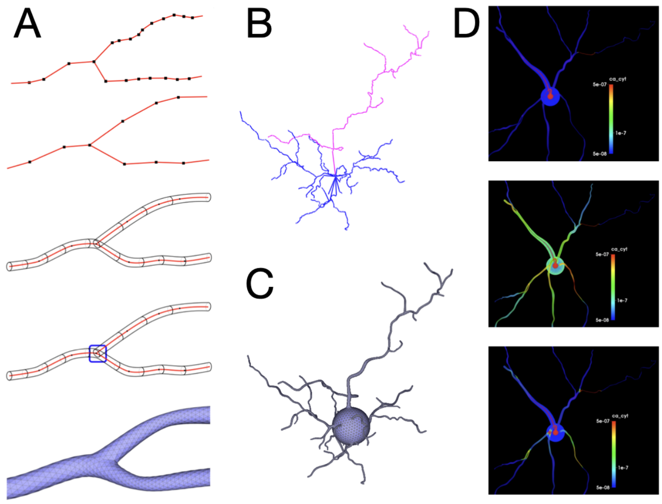


Figure 3: **3D reconstruction pipeline**: (A) Illustration of cell surface reconstruction using graph and diameter information available from NeuroMorpho.org. The reconstruction pipeline begins with cellular graph information (B) and constructs a watertight surface triangulation (C), using the method presented in (?). (D) Time evolution of a whole cell calcium simulation.

Numerical method: The numerical method for this sub-project is identical to Sec. **??**.
Estimate of proposed simulation runs: To study the cellular calcium dynamics from synapses to the cell nucleus, three computational setups will be used and solved on SDSC Comet. Representative computational meshes, which will be used in the studies outlined below and used in the strong and weak scaling benchmarks have roughly 451,485 DoFs (see Tab. **??**, **??**) with run times of roughly 5 minutes for 1,000 timestep on 3,074 processors.

1. `Variation of location of synaptic stimulation` – We will use 10 neurons from the NeuroMorpho.org database and generate the 3D cell and ER geometries using the approach outlined above. Then we will vary the location of 500 synapses on the dendrites to determine the conditions under which calcium waves towards the nucleus are triggered.

   For these simulation setups a simulation time of 2 seconds is necessary to track calcium waves through the entire neuron. With a timestep = 0.1 ms this will amount to $10 \times \frac{2{,}000}{0.1} = 200,000$ timesteps for one specific synaptic stimulation pattern. To address the question how spatial variation of synaptic stimulation will affect calcium waves, 5 distribution patterns will be tested: homogeneously distributed synapses 1) close to soma, 2) on distal dendrites, 3) on proximal dendrites, and clustered synapses (non-homogeneous) 4) on distal dendrites, and 5) proximal dendrites. In total $5 \times 10 \times \frac{2{,}000}{0.1} = 1,000,000$ timesteps will be necessary. Using the benchmark data above, this yields $\frac{1{,}000{,}000}{1{,}000} \times \frac{5}{60} \times 3072 = 256,000$ core hours.

2. `Variation of synaptic activation strength` – Using the same 10 neurons we will vary synaptic strength to determine the conditions under which stable synapse-to-nucleus calcium communication is observed. For this study a simulation time of 2 seconds is also necessary. We will test 3 settings: Low, intermediate and high synapse strength. Thus, in total we expect $3 \times 10 \times \frac{2{,}000}{0.1} = 600,000$ time steps, which yields $\frac{600{,}000}{1{,}000} \times \frac{5}{60} \times 3,072 = 153,600$ core hours.

3. `Synapse loss study` – We will employ 6 representative cells to study the effect of synapse loss on synapse-to-nucleus calcium communication. Comparing the healthy state (all synapses active) to pathological states in which we iteratively decrease the active synapse population by 10 %, we will determine at which pathological states communication break-down is observed. Abortive calcium signals become visible in the first two seconds, thus we will set our end time to 2 s. Using a time step = 0.1 ms to resolve action potentials $20,000$ time steps are required for one neuron. In total, for 6 neurons with 10 states (1 healthy, 9 pathological) and 20,000 time steps for each simulation a total of $6 \times 10 \times \frac{2{,}000}{0.1} = 1,200,000$ time steps is required. Using the above benchmark data yields $\frac{1{,}200{,}000}{1{,}000} \times \frac{5}{60} \times 3,072 = 307,200$ core hours.

For a summary of representative simulation runs, see Table **??** with anticipated compute time (core-h / SUs) requirements for this sub-project. Storage requirement for one VTK file is roughly 1 MiB per time step. This yields, for all three studies, a total of 2.8 TiB.

4.3 Summary

The resources required for the two projects outlined in Sec. **??** and **??** are summarized in Tab. **??**. The graduate student James Rosado will be conducting all runs related to Sec. **??** and Stephan Grein will be responsible for all runs for Sec. **??**. The postdoc Qingguang Guan and PI Queisser will support and advise efforts in both projects.

All participants in the project will have access to a shared file in which an estimated demand for compute time for the following project month can be submitted. Stephan Grein will coordinate the internal distribution of compute time for the subsequent month by weighing the individual demand against milestone deadlines, demand for conference contributions, theses finalizations and other constraints. Thus, contingent overflow as well as monthly over-demand can be minimized.

5 Justification of Resources Requested

The charge unit for SDSC Comet is the Service Unit (SU). One SU corresponds to the use of one compute core for one hour (The minimum charge for a job longer than 10 seconds is 1 SU). Accordingly the requested SUs for the sub-projects in this proposal are summarized in Tab. **??**, which have been determined by the following equation:

$$SUs = \sum Runtime(Walltime)\ [h] \quad \times \quad \#\ Cores \quad \times \quad \#\ Runs$$

The total storage requirement for the two sub-projects is **4.8 TiB**, in which the sub-project **??** requires **2 TiB** and sub-project **??** requires **2.8 TiB** of storage.

Compute time requirements for **SDSC Comet**						
Sub-project	Type of run	# runs	# Simulation Time /run [sec.]	Wall time /run [h]	# cores/ run	Total [core-h] (SU)
Synaptic spine dynamics	Baseline	90	0.02	1.00	30	2,700
Synaptic spine dynamics	RyR	9000	0.02	1.00	30	270,000
Synaptic spine dynamics	Calcium pulse	135	3.00	1.00	4,500	607,500
Whole cell dynamics	Synaptic location	50	2.00	$1.\overline{66}$	3,072	256,000
Whole cell dynamics	Synaptic strength	30	2.00	$1.\overline{66}$	3,072	153,600
Whole cell dynamics	Synapse loss	60	2.00	$1.\overline{66}$	3,072	307,200
TOTAL						**1,597,000**

Table 1: Total number of anticipated SUs on SDSC Comet for the two requested sub-projects as outlined in sections **??**, **??**.

6 Additional considerations

Dr. Queisser (PI) and his research group has a long-standing experience in Scientific Computing. Prior to Dr. Queisser's relocation to the US from Germany, his group was awarded resources on the JUQUEEN and JUWELS supercomputers in Jülich, Germany over the period of 6 consecutive years. Currently, Dr. Queisser's postdoctoral researcher Dr. Guan and graduate student James Rosado are actively working on the subproject **??** which is funded through a recently awarded NIH grant. Both have experience in running code on remote parallel compute clusters. Furthermore, graduate student Stephan Grein is involved in the subproject **??**. James Rosado and Stephan Grein ran the presented XSEDE benchmarks. The requested SU are relevant for carrying out the NIH project and for the completion of Grein's dissertational research. This project will support 70% of the National Health Institute (NIH) grant *CRCNS: Collaboration toward an experimentally validated multiscale model of rTMS*. The remaining 30% are requested to support Grein's dissertation completion. Our group has access to the smaller-sized parallel computing system Owlsnest operated by Temple University, Philadelphia, with roughly 180 general computing CPUs plus some GPU capabilities on other nodes. While developing on local multicore machines, then porting to the local compute cluster Owlsnest for testing, debugging and running smaller batch parameter sweeps is feasible on the local infrastructure, we require access to larger machines for running production code simulations to validate publishable research results. These simulations are beyond what we can run locally under the given constraints.